X(3872) Analysis Update

new findings in reweighting and ML

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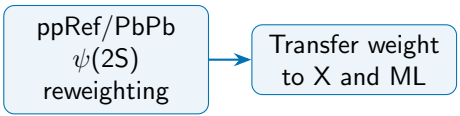






Recap: two routes we tried

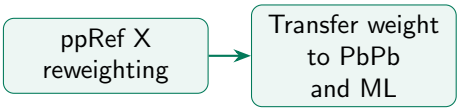
ψ(2S) ppRef/PbPb weight $\rightarrow$ X



What stopped this route

- ψ(2S) and X MC have different variable distributions.
- In PbPb, no sufficiently significant ψ(2S) peak was available for reweighting.

X ppRef weight $\rightarrow$ PbPb



What stopped this route

- X ppRef and PbPb MC showed different input distributions.
- No visible X peak remained after the transfer and ML selection.

How the present strategy emerged

1. Try the ψ(2S) route

Transfer the ppRef $\psi(2S)$ weight to PbPb and ML.

Tighten the control selection and inspect the X mass.

No X peak found.

2. Find a pre-cut mismatch

ppRef:
`BtrkPt`$>$ 0.5 GeV

PbPb:
`BtrkPt`$>$ 0.9 GeV

Earlier reweighting used only the ppRef fiducial region.

3. Use a common fiducial region

Require
`BtrkPt`$>$ 0.9 GeV
in both systems.

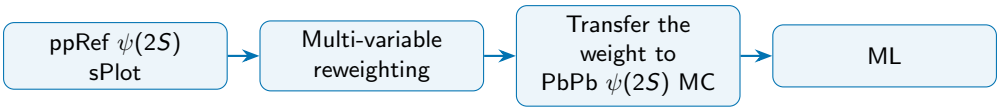
Repeat the MC comparison, X ppRef reweighting and ML.

4. Tighten cuts

Apply a `Btrk2dR`$<$ 0.25 in ML since we can't use both BrkdR in ML.

A hint of X in pbpb appears.

$\psi(2S)$ ppRef weight $\rightarrow$ PbPb workflow



Common reweighting setting

Variables: [Bcos_dtheta, Btktkpt, Bchi2Prob, Btrk2Pt, Btrk1Pt]

$10 <$ Bpt < 50 GeV, $|$By$| < 1.6$, BQvalue < 0.15, $2 <$ Btktkpt < 10 GeV, Btrk1dR < 0.5

R5range2

BtrkPt> 0.5 GeV

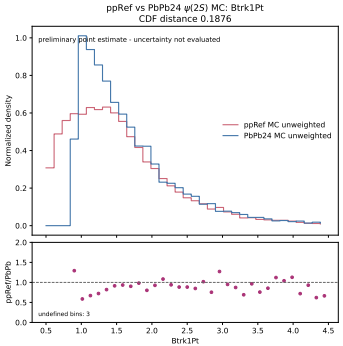
R5range3

BtrkPt> 1.0 GeV

Flat-tree pre-cuts: ppRef BtrkPt> 0.5 GeV; PbPb BtrkPt> 0.9 GeV.

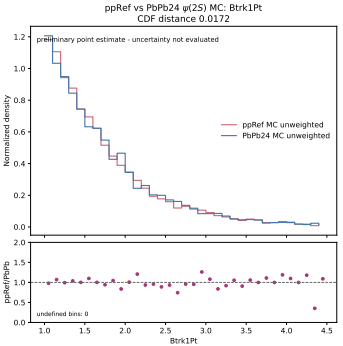
Aligning the BtrkPt pre-cut aligns the MC domains

R5range2: effective pre-cuts differ



KS-like distance = 0.1876
R8 mean = 0.10709

R5range3: common pre-cut



KS-like distance = 0.0172
R8 mean = 0.01721

With the same BtrkPt pre-cut, ppRef and PbPb $\psi(2S)$ MC are very close.

ML settings based on the same reweighting R5range3

Weighted ML baseline

$10 < \texttt{Bpt} < 50$ GeV, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$, $\texttt{BtrkPt} > 1.0$ GeV, $2 < \texttt{Btktkpt} < 10$ GeV, $\texttt{Btrk1dR} < 0.5$

Variables: [Bchi2Prob, Btrk1dR, BtrkPtimb, Btrk1Pt, Btrk2Pt, BtktkvProb, Bcos_dtheta, Btktkpt]

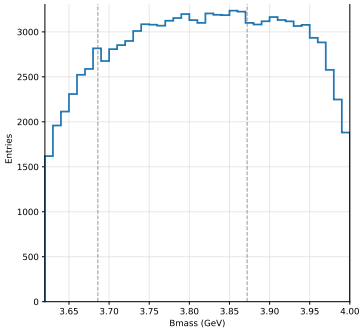
Baseline tag: X_pb24_v7_fid8_8v2_rwpsi2sr5range3v1_xgb_v1

Train tag	Difference from the weighted 8-variable baseline
X_pb24_v7_fid8_8v2_rwpsi2srawv1_xgb_v1	Use raw $\psi(2S)$ PbPb MC without weight
X_pb24_v7_fid8_6v5_rwpsi2sr5range3v1_xgb_v1	Remove BtrkPtimb and BtktkvProb
X_pb24_v7_fid8_7v2_rwpsi2sr5range3v1_xgb_v1	Add Btrk2dR; remove Btrk2Pt and BtktkvProb
X_pb24_v8_fid9_8v2_rwpsi2sr5range3dr045v1_xgb_v1	Add Btrk2dR<0.45 selection

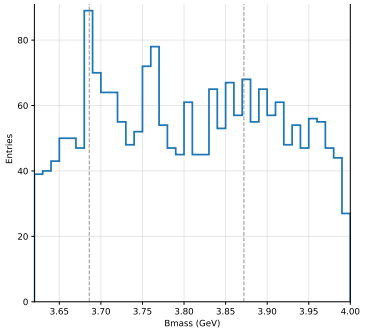
The $\psi(2S)$ control improves, but X remains absent

Weighted ML baseline; score cuts are defined by weighted $\psi(2S)$ efficiency.

Loose ML selection: 80% efficiency



Tight ML selection: 20% efficiency



Across the five ML settings, tighter selections enhance the $\psi(2S)$ peak, but no clear narrow X peak is visible.

Why return to X ppRef weight → PbPb?

What the $\psi(2S)$ route established

- Common `BtrkPt` pre-cuts bring ppRef and PbPb $\psi(2S)$ MC close.
- ML selections produce a stronger $\psi(2S)$ control peak.

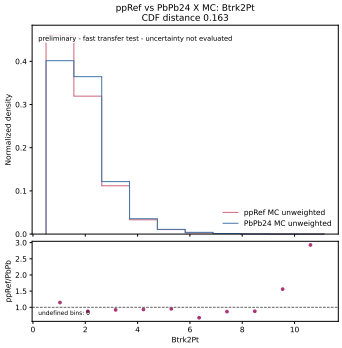
What it did not deliver

- None of the five ML settings produced a clear narrow X peak.
- The optimization of ML performance was not studied in detail.

Align the X `BtrkPt` pre-cuts → Repeat X ppRef weight → PbPb

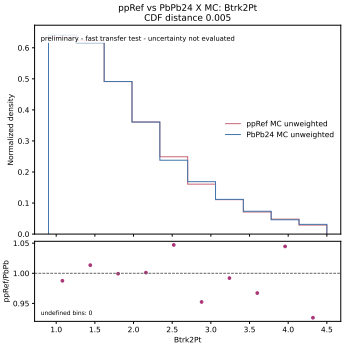
Aligning the BtrkPt pre-cut aligns the X MC domains

Original: different pre-cuts



KS-like distance = 0.1626
R8 mean = 0.09413

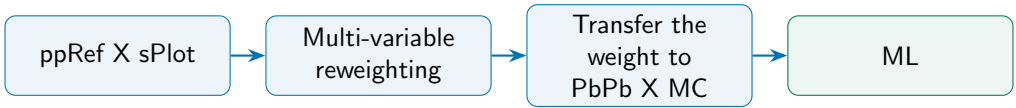
Same pre-cut: BtrkPt> 0.9 **GeV**



KS-like distance = 0.0049
R8 mean = 0.01307

With the same BtrkPt pre-cut, ppRef and PbPb X MC are very close.

X ppRef weight → PbPb: workflow



What changes were tried

- Same pre-cut
- Different reweighting variables & fiducial regions
- Different ML variables & fiducial regions

What improved

- ppRef/PbPb MC agreement in the same fiducial region
- ML-to-fit workflow
- A hint of X in PbPb

Three X ppRef reweighting settings and ML follow-ups

Common variables: [Bcos_dtheta, Btktkpt, Bchi2Prob, Btrk1Pt, Btrk1dR]

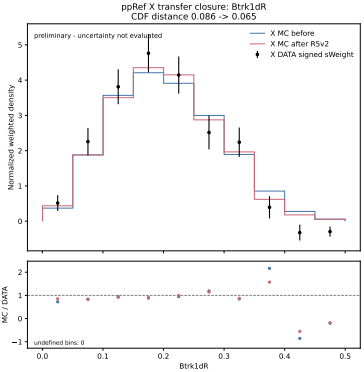
Common region: $10 < \texttt{Bpt} < 50$ GeV, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$, $2 < \texttt{Btktkpt} < 8$ GeV, $\texttt{BtrkPt} > 0.9$ GeV

Setting	BDT reweighting variables and region	ML follow-up
R5v2range3	Variables: common set Region: $\texttt{Btrk1Pt} \leq 4.5$ GeV	No ML follow-up
R6range3	Variables: common + Btrk2Pt Region: $\texttt{BtrkPt} \leq 4.5$ GeV	8v2: [Bchi2Prob, Btrk1dR, BtrkPtimb, Btrk1Pt, Btrk2Pt, BtktkvProb, Bcos_dtheta, Btktkpt] 6v5: [Btrk1dR, Btktkpt, Btrk1Pt, Bchi2Prob, Bcos_dtheta, Btrk2Pt] 6v5 regions: $\texttt{Btrk2dR} < 0.45$ and < 0.25
R6v2range3	Variables: common + Btrk2dR Region: $\texttt{Btrk2dR} < 0.5$	8v2: same variable set as above 9v2: [Btrk1dR, Btrk2dR, Btktkpt, Btrk1Pt, Btrk2Pt, Bchi2Prob, Bcos_dtheta, BtrkPtimb, BtktkvProb]

Different reweighting variables were tested to find the most suitable variable set.

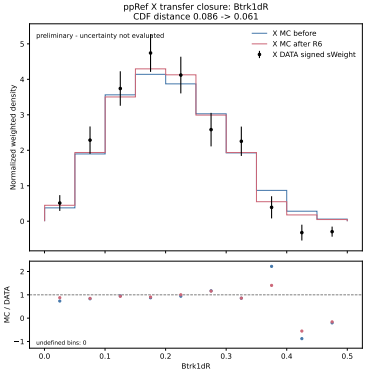
Btrk1dR sWeighted-data closure

R5v2range3



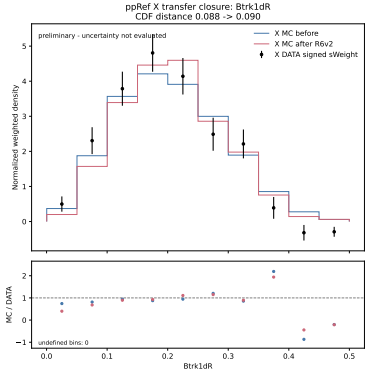
KS-like distance
Btrk1dR: 0.0856 → 0.0646
Mean: 0.1403 → 0.0410

R6range3



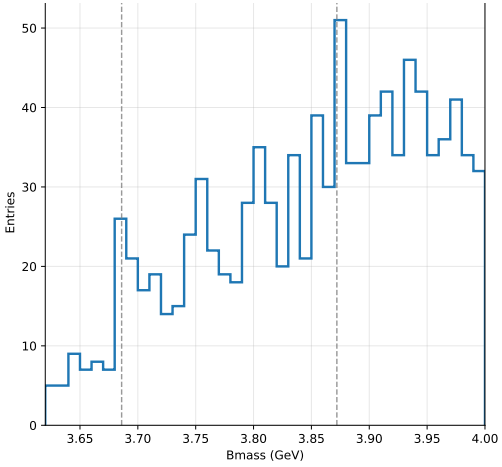
KS-like distance
Btrk1dR: 0.0860 → 0.0612
Mean: 0.1350 → 0.0395

R6v2range3



KS-like distance
Btrk1dR: 0.0879 → 0.0901
Mean: 0.1423 → 0.0673

A pronounced peak-like feature at xeff 20%



Setting

- R6range3 weight
- ML variables: [`Btrk1dR`, `Btktkpt`, `Btrk1Pt`, `Bchi2Prob`, `Bcos_dtheta`, `Btrk2Pt`]
- ML fiducial region: `Btrk2dR`< 0.25
- Weighted signal efficiency: 20%

Unlike the previous ML settings, a pronounced local feature appears near the X mass.

Mass-spectrum observation before a dedicated fit interpretation.

Scanning the Btrk2dR region in reweighting and ML

Common setting

Reweighting variables: [Bcos_dtheta, Btktkpt, Bchi2Prob, Btrk2Pt, Btrk1Pt, Btrk1dR]

ML variables: [Btrk1dR, Btktkpt, Btrk1Pt, Bchi2Prob, Bcos_dtheta, Btrk2Pt]

Reweighting setting	Reweighting fiducial region	ML setting	ML fiducial region
R6range4	Btrk2dR<0.45	6v5 with R6range4 weight	Btrk2dR<0.45
R6range5	Btrk2dR<0.40	6v5 with R6range5 weight	Btrk2dR<0.40
R6range6	Btrk2dR<0.35	6v5 with R6range6 weight	Btrk2dR<0.35
R6range7	Btrk2dR<0.30	6v5 with R6range7 weight	Btrk2dR<0.30
R6range8	Btrk2dR<0.25	6v5 with R6range8 weight	Btrk2dR<0.25

For each setting, reweighting and ML use the same Btrk2dR fiducial region.

Nominal-fit scan across the five settings

DATA-only nominal fit: extended unbinned fit in $3.80 < \texttt{Bmass} < 3.94$ GeV; single Gaussian, $3.86 < \mu < 3.88$ GeV, $0.1 < \sigma < 10$ MeV; Chebyshev-2 background.

xeff scan: strict `Prediction`>threshold at weighted signal efficiency $= 10, 15, 20, 25, 30, 35, 40\%$.

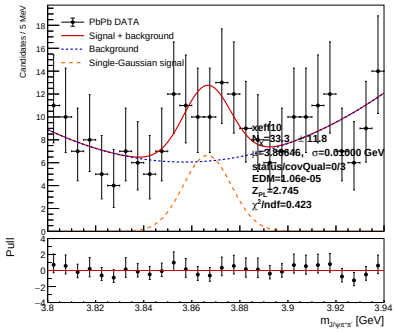
Each cell: local significance σ / fitted yield $\pm$ stat.

Setting \ xeff	10%	15%	20%	25%	30%	35%	40%
R6range4	2.11σ / 33.4 ± 16.1	1.42σ / 36.4 ± 25.2	0.00σ /[B] 0.1 ± 239.7	0.55σ / 15.7 ± 34.2	1.56σ /[B] 56.6 ± 44.5	1.39σ / 65.5 ± 51.9	1.08σ / 61.0 ± 62.9
R6range5	1.92σ /[B] 29.1 ± 15.0	2.09σ / 42.0 ± 30.4	1.37σ / 44.1 ± 46.4	1.15σ / 31.9 ± 33.0	0.93σ /[B] 23.1 ± 28.4	1.57σ / 68.7 ± 48.2	1.55σ / 80.6 ± 58.6
R6range6	2.00σ / 27.5 ± 13.5	1.82σ /[B] 40.6 ± 22.2	1.13σ /[B] 35.0 ± 30.7	0.76σ / 29.6 ± 36.7	1.28σ /[B] 64.7 ± 51.3	1.55σ / 55.2 ± 42.6	1.33σ / 52.7 ± 53.2
R6range7	2.75σ /[B] 33.3 ± 11.8	2.09σ / 16.6 ± 10.7	1.57σ /[B] 40.4 ± 25.6	1.01σ /[B] 32.6 ± 32.1	0.74σ /[B] 30.3 ± 40.6	2.02σ / 41.3 ± 24.6	0.04σ /[F] 2.7 ± 942.7
R6range8	1.99σ /[B] 19.8 ± 9.6	2.07σ / 12.2 ± 7.5	1.60σ /[B] 32.2 ± 19.9	1.81σ / 27.0 ± 18.1	1.45σ / 32.3 ± 28.6	0.36σ / 13.6 ± 37.4	0.00σ /[B] 0.0 ± 48.5

[B] parameter boundary reported [F] fit status/covQual $\neq 0/3$

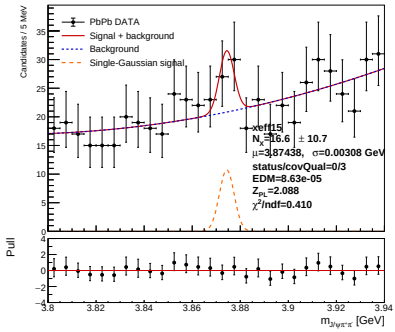
R6range7 nominal fits at xeff 10% and 15%

xeff 10%



Local significance: 2.745σ
Yield: 33.26 ± 11.80 (stat.)
Gaussian width: 10.000 MeV [boundary]

xeff 15%



Local significance: 2.088σ
Yield: 16.55 ± 10.66 (stat.)
Gaussian width: 3.079 MeV

Fixing the ML Btrk2dR region while varying reweighting

Common setting

Reweighting variables: [Bcos_dtheta, Btktkpt, Bchi2Prob, Btrk2Pt, Btrk1Pt, Btrk1dR]

ML variables: [Btrk1dR, Btktkpt, Btrk1Pt, Bchi2Prob, Bcos_dtheta, Btrk2Pt]

Reweighting setting	Reweighting fiducial region	ML setting	ML fiducial region
R6range4	Btrk2dR<0.45	6v5 with R6range4 weight	Btrk2dR<0.25
R6range5	Btrk2dR<0.40	6v5 with R6range5 weight	Btrk2dR<0.25
R6range6	Btrk2dR<0.35	6v5 with R6range6 weight	Btrk2dR<0.25
R6range7	Btrk2dR<0.30	6v5 with R6range7 weight	Btrk2dR<0.25

The ML fiducial region is fixed; only the transferred weight setting changes.

Nominal-fit scan with a common ML Btrk2dR region

DATA-only nominal fit: extended unbinned fit in $3.80 < \texttt{Bmass} < 3.94$ GeV; single Gaussian, $3.86 < \mu < 3.88$ GeV, $0.1 < \sigma < 10$ MeV; Chebyshev-2 background.

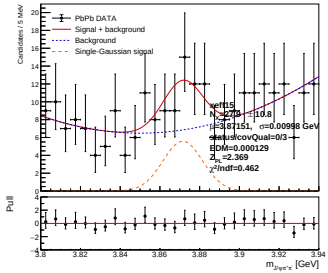
xeff scan: strict Prediction>threshold at weighted signal efficiency $= 10, 15, 20, 25, 30, 35, 40\%$.

Each cell: local significance σ / fitted yield $\pm$ stat.

Setting \ xeff	10%	15%	20%	25%	30%	35%	40%
R6range4	1.67σ / 7.5 ± 5.7	2.37σ / 27.9 ± 10.8	2.50σ /[B] 42.7 ± 17.3	1.40σ / 22.6 ± 19.2	2.14σ / 53.1 ± 40.8	1.05σ / 19.8 ± 21.8	2.65σ / 13.7 ± 6.5
R6range5	2.24σ / 18.7 ± 8.1	2.24σ / 26.7 ± 11.8	1.82σ / 30.5 ± 20.1	2.33σ / 35.3 ± 18.2	1.68σ / 33.8 ± 26.6	1.27σ / 16.9 ± 17.3	1.36σ / 25.3 ± 28.6
R6range6	2.15σ / 18.5 ± 10.1	2.35σ / 29.4 ± 12.5	2.48σ /[B] 44.5 ± 17.9	1.74σ / 26.3 ± 18.9	1.66σ / 22.6 ± 18.4	0.19σ / 5.8 ± 37.7	0.83σ /[F] 14.5 ± 23.3
R6range7	2.16σ /[B] 19.8 ± 8.9	2.69σ /[B] 36.3 ± 13.2	1.30σ / 23.8 ± 18.1	2.13σ / 36.3 ± 22.7	1.46σ / 22.0 ± 17.5	0.77σ / 15.6 ± 22.3	1.50σ /[B] 4.8 ± 3.7

[B] parameter boundary reported [F] fit status/covQual $\neq 0/3$

Several best ML results

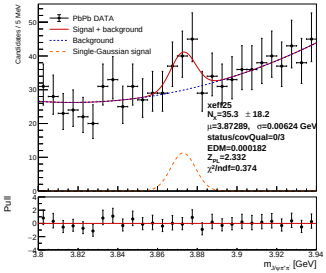


R6range4 · xeff 15%

Local significance: 2.369σ

Yield: 27.89 ± 10.82 (stat.)

Gaussian width: 9.982 MeV

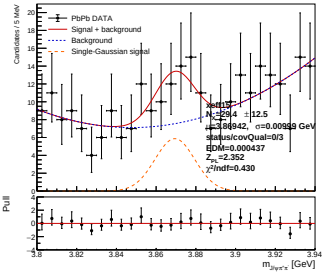


R6range5 · xeff 25%

Local significance: 2.332σ

Yield: 35.28 ± 18.21 (stat.)

Gaussian width: 6.237 MeV



R6range6 · xeff 15%

Local significance: 2.352σ

Yield: 29.37 ± 12.48 (stat.)

Gaussian width: 9.988 MeV

Train-wide and application-tight test

Common setting

R6range5 weight; 6v5 ML variables; training region: `Btrk1dR`<0.45, `Btrk2dR`<0.40; same XGBoost model

Setting	Training region	Application/fit region	Train/test AUC
Baseline	`Btrk2dR`<0.40	`Btrk2dR`<0.40	0.88334 / 0.86113
Train-wide/apply-tight	`Btrk2dR`<0.40	`Btrk2dR`<0.25	same model

Fit change for the new setting: Gaussian width range $[0.1, 10] \rightarrow [2, 8]$ MeV.

Each cell: local significance σ / fitted yield $\pm$ stat.

Setting \ xeff	10%	15%	20%	25%	30%	35%	40%
Baseline	1.92σ /[B] 29.1 ± 15.0	2.09σ / 42.0 ± 30.4	1.37σ / 44.1 ± 46.4	1.15σ / 31.9 ± 33.0	0.93σ /[B] 23.1 ± 28.4	1.57σ / 68.7 ± 48.2	1.55σ / 80.6 ± 58.6
Apply-tight	2.23σ / 14.5 ± 6.7	2.09σ /[B] 20.1 ± 10.0	1.98σ /[B] 28.5 ± 14.6	2.23σ / 40.3 ± 22.1	1.78σ /[B] 41.4 ± 22.7	1.76σ / 50.4 ± 30.6	1.31σ / 26.0 ± 25.7

[B] parameter boundary reported New setting: same training and model, tighter application/fit region.

Narrower background mass region test

Common setting: R6range5 weight; 6v5 ML variables; ML fiducial region: `Btrk1dR`<0.45, `Btrk2dR`<0.25

Setting	Background mass region [GeV]	XGBoost model	Train/test AUC
Baseline	$[3.75, 3.80] \cup [3.95, 4.00]$	baseline	0.93936 / 0.92585
Narrow background	$[3.82, 3.85] \cup [3.89, 3.92]$	retrained	0.91844 / 0.89587

Fit change for the new setting: Gaussian width range $[0.1, 10] \rightarrow [2, 8]$ MeV.

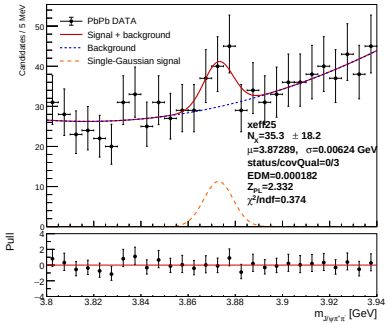
Each cell: local significance σ / fitted yield $\pm$ stat.

Setting \ xeff	10%	15%	20%	25%	30%	35%	40%
Baseline	2.24σ / 18.7 ± 8.1	2.24σ / 26.7 ± 11.8	1.82σ / 30.5 ± 20.1	2.33σ / 35.3 ± 18.2	1.68σ / 33.8 ± 26.6	1.27σ / 16.9 ± 17.3	1.36σ / 25.3 ± 28.6
Narrow background	2.18σ /[B] 5.9 ± 3.3	3.08σ /[B] 31.0 ± 10.5	3.22σ /[B] 45.0 ± 14.5	3.25σ /[B] 58.6 ± 18.7	2.21σ /[B] 49.7 ± 23.0	1.93σ / 53.3 ± 27.8	1.92σ / 62.7 ± 33.5

[B] parameter boundary reported

Background region and XGBoost model both change.

Best ML result comparison

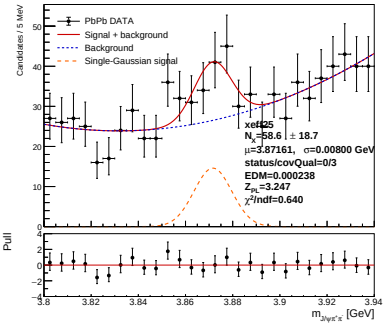


Baseline · xeff 25%

Local significance: 2.332σ

Yield: 35.28 ± 18.21 (stat.)

Gaussian width: 6.237 MeV



Narrow background · xeff 25%

Local significance: 3.247σ

Yield: 58.55 ± 18.68 (stat.)

Gaussian width: 7.999 MeV [boundary]

Summary and Outlook

Summary

- The $\psi(2S)$ ppRef weight $\rightarrow$ PbPb workflow shows no clear X peak.
- With the same pre-cut, a tight `Btrk2dR` selection, and sidebands closer to the X mass region, the X ppRef weight $\rightarrow$ PbPb workflow shows a good hint of an X peak in PbPb.
- Preliminary but exciting progress; non-smooth mass spectra and a small, unstable peak remain unresolved.

Outlook

- Complete the $\psi(2S)$ ppRef weight $\rightarrow$ PbPb workflow: use PbPb $\psi(2S)$ sWeighted data to evaluate the reweighted PbPb MC variables, then apply the validated workflow to X ppRef weight $\rightarrow$ PbPb.
- Test different reweighting, ML, and application variables and fiducial regions.

Thanks for listening!

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